EXPERIMENT 18: Producer-Consumer Problem Using Semaphores

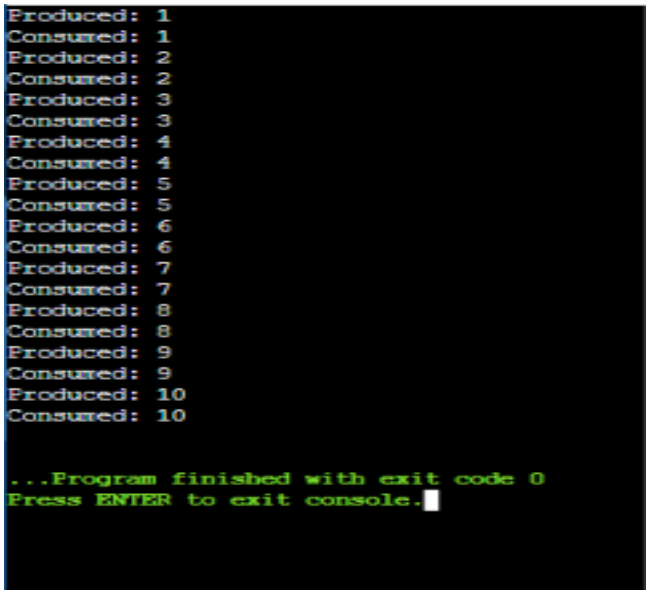
code:

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

#define SIZE 5
#define ITEMS 10

int buffer[SIZE];
int in = 0, out = 0;

sem_t empty, full;
pthread_mutex_t mutex;

void *producer(void *arg) {

   for (int i = 1; i <= ITEMS; i++) {

      sem_wait(&empty);
      pthread_mutex_lock(&mutex);

      buffer[in] = i;

      printf("Produced: %d\n", i);

      in = (in + 1) % SIZE;
```

```
        pthread_mutex_unlock(&mutex);
        sem_post(&full);

        sleep(1);
    }

    return NULL;
}

void *consumer(void *arg) {

    for (int i = 1; i <= ITEMS; i++) {

        sem_wait(&full);
        pthread_mutex_lock(&mutex);

        int item = buffer[out];

        printf("Consumed: %d\n", item);

        out = (out + 1) % SIZE;

        pthread_mutex_unlock(&mutex);
        sem_post(&empty);

        sleep(1);
    }

    return NULL;
}
```

```
int main() {

    pthread_t p, c;

    sem_init(&empty, 0, SIZE);
    sem_init(&full, 0, 0);

    pthread_mutex_init(&mutex, NULL);

    pthread_create(&p, NULL, producer, NULL);
    pthread_create(&c, NULL, consumer, NULL);

    pthread_join(p, NULL);
    pthread_join(c, NULL);

    sem_destroy(&empty);
    sem_destroy(&full);

    pthread_mutex_destroy(&mutex);

    return 0;
}
```

Output:


Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Consumed: 3
Produced: 4
Consumed: 4
Produced: 5
Consumed: 5
Produced: 6
Consumed: 6
Produced: 7
Consumed: 7
Produced: 8
Consumed: 8
Produced: 9
Consumed: 9
Produced: 10
Consumed: 10
...Program finished with exit code 0
Press ENTER to exit console.